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Expression profile of the entire detoxification gene inventory of the western honeybee, *Apis mellifera* across life stages

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A B S T R A C T

The western honeybee, *Apis mellifera,* is a managed pollinator of many crops and potentially exposed to a wide range of foreign compounds, including pesticides throughout its life cycle. Honeybees as well as other insects recruit molecular defense mechanisms to facilitate the detoxification of xenobiotic compounds. The inventory of detoxification genes (DETOXome) is comprised of five protein superfamilies: cytochrome P450 monooxygenases (P450), carboxylesterases, glutathione S-transferases (GST), UDP-glycosyl transferases (UGT) and ATP-binding cassette (ABC) transporters. Here we characterized the gene expression profile of the entire honeybee DETOXome by analyzing 47 transcriptomes across the honeybee life cycle, including different larval instars, pupae, and adults. All life stages were well separated by principal component analysis, and K-means clustering revealed distinct temporal patterns of gene expression. Indeed, >50% of the honeybee detoxification gene inventory is found in one cluster and follows strikingly similar expression profiles, i.e., increased expression during larval development, followed by a sharp decline after pupation and a steep increase again in adults. This cluster includes 29 P450 genes dominated by CYP3 and CYP4 clan members, 15 ABC transporter genes mostly belonging to the ABCC subfamily and 13 carboxylesterase genes including almost all members involved in dietary/detox and hormone/semiochemical processing. RT-qPCR analysis of selected detoxification genes from all families revealed high expression levels in various tissues, especially Malpighian tubules, fatbody and midgut, supporting the view that these tissues are essential for metabolic clearance of environmental toxins and pollutants in honeybees. Our study is meant to spark further research on the molecular basis of detoxification in this critical pollinator to better understand and evaluate negative impacts from potentially toxic substances. Additionally, the entire gene set of 47 transcriptomes collected and analyzed provides a valuable resource for future honeybee research across different disciplines.

1. Introduction

Plant pollination by insect pollinators is an indispensable ecosystem service (Klein et al., 2006; Ollerton et al., 2011), including crop pollination in agricultural production systems by commercially managed bee pollinators such as the western honey bee, *Apis mellifera* (Potts et al., 2016). Honeybees are likely to be exposed to a broad range of xenobiotic compounds including pesticides while foraging in agricultural cropping systems (Johnson, 2015). Especially insecticides have the potential to be acutely toxic to honeybees, particularly those addressing phylogenetically conserved target sites among insects such as ligand- and voltage-gated ion channels (Casida and Durkin, 2013). To overcome toxic effects by foreign compounds, insects including honeybees recruit enzyme-based defense mechanisms mediating their detoxification (Johnson, 2015). A milestone for honeybee research was the publication of the honeybee genome in 2006 (Weinstock et al., 2006), and the subsequent assembly and annotation of detoxification gene superfamilies involved in the metabolism and clearance of xenobiotics (Claudianos et al., 2006). Metabolism of exogenous compounds can be divided into different phases. In phase I, compounds are functionalized to often less toxic and/or easier excretable metabolites upon conjugation (Amezian et al., 2021; Yu, 2011). Major enzyme superfamilies involved in phase I are cytochrome P450 monooxygenases (P450s) and carboxyl/cholinesterases (CCEs). In phase II, metabolites from phase I or the parent compound itself are conjugated with endogenous molecules such as glutathione or sugars further decreasing their biological activity while increasing water solubility. Essential enzyme families facilitating phase II metabolic fate of xenobiotic compounds in insects are glutathione S-

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transferases (GSTs) and uridine diphosphate (UDP)-glycosyltransferases (UGTs). Phase III describes the export / elimination of compounds and their metabolites from cells or tissues via transporter systems such as the ATP-binding cassette (ABC) transporters. Members of all these enzyme superfamilies have been implicated in phenotypic insecticide resistance in a vast array of insect pest species highlighting their relevance for insecticide detoxification and tolerance in insects in general (for reviews see Dermauw and Van Leeuwen, 2014; Enayati et al., 2005; Koirala et al., 2022; Montella et al., 2012; Nagare et al., 2021; Nauen et al., 2022).

Initial concerns that a deficit in the detoxification gene inventory (DETOXome) in honeybees translates to higher sensitivity towards insecticides in general (Claudianos et al., 2006) was not supported by empirical data (Hardstone and Scott, 2010). However, it was found that most major chemical classes of insecticides such as pyrethroids, organophosphates and neonicotinoids have members which exhibit high toxicity to honeybees while others are only of moderate or low acute toxicity (Arena and Sgolastra, 2014). These cases of differential sensitivity are often related to toxicokinetic properties, most importantly the differential capacity to metabolize and detoxify these compounds.

Especially CYP3 clan P450 families (CYP6 and CYP9 enzymes) have been shown to be molecular determinants of selectivity resulting in a marked tolerance against different insecticides such as the pyrethroid *tau*-fluvalinate (Mao et al., 2011), the organophosphate coumaphos (Mao et al., 2011), the neonicotinoids thiacloprid and acetamiprid (Manjon et al., 2018), the butenolide flupyradifurone (Haas et al., 2021b), the diamide chlorantraniliprole (Haas et al., 2021a) as well as natural plant-derived toxins (Mao et al., 2009; du Rand et al., 2015). In addition, to their general ability to metabolize these compounds, the spatiotemporal expression pattern of P450s is a further indication of their predominant role in xenobiotic metabolism. Candidate P450s such as CYP9Q3 have been found upregulated in late larval stages and adult workers (Haas et al., 2021b), and shown to be highly expressed in tissues involved in xenobiotic detoxification such as the Malpighian tubules (Manjon et al., 2018). Other studies revealed high expression levels in appendages such as legs and antennae primarily exposed to foreign compounds by contact (Mao et al., 2015).

While CYP3 clan P450s have been investigated thoroughly regarding their role in insecticide toxicology in honeybees, other enzyme superfamilies have not been studied in detail and their role remains largely elusive. Synergist studies suggest a (minor) involvement of CCEs and ABC transporters in insecticide metabolism (Hawthorne and Dively, 2011; Johnson et al., 2006) and several studies reported elevated enzyme activities or transcriptional upregulation after insecticide exposure (Berenbaum and Johnson, 2015). But no study to the best of our knowledge has provided a direct, functional link between a specific gene and insecticide metabolism from any of the other major detoxification gene superfamilies (CCEs, GSTs, UGTs or ABCs) in honeybees. Based on the honeybee's life cycle, environmental response genes are assumed to be expressed primarily in adults foraging and processing food, but also later larval stages no longer feeding on pre-processed diet from nurse bees, but nectar and pollen.

To identify promising candidates out of detoxification enzyme families other than P450s and likely to be involved in xenobiotic clearance in the honeybee, we analyzed patterns of expression of the entire honeybee detoxification gene inventory based on a previously published review (Berenbaum and Johnson, 2015). We analyzed transcript levels of the honeybee DETOXome across different life stages by sequencing 47 transcriptomes and employed K-means clustering to broadly cluster genes according to their expression profile. The overarching objective of the study was to track the gene expression profile of the honeybee DETOXome and to identify those genes out of this inventory following temporal expression patterns suggesting an involvement in xenobiotic detoxification. In addition, we think that the global gene set of 47 transcriptomes collected and analyzed provides a valuable resource to inform current and future research projects on honeybees and beyond.

2. Material and methods

2.1. Sample collection

Honeybees (*Apis mellifera*) of different life stages were collected from queen-right colonies in Monheim am Rhein, Germany in 2016. The colonies had not received any chemical treatment for at least 6 months and their health status was checked weekly by visual inspection. To ensure synchronization of bees, the queen of each hive was confined in their own colony in an exclusion cage containing an empty comb for 24 h. After this egg laying period, the queen was released, and the comb was put back in the exclusion cage and into the colony. After different time intervals samples were taken. Five different time points were sampled, including larvae (4 days after oviposition (DAO), 6 DAO, 8 DAO and 11 DAO) and pupae (18 DAO). Additionally, adult bees, separated according to their collection site (nurse: brood chamber; worker: honey chamber; forager: flight hole) were collected resulting in a total of eight different life stages (Fig. 1). Biological replicates comprising 5 individuals per replicate were collected from three different hives at two sampling dates for each hive between the end of May and mid-July (resulting in 6 biological replicates for each developmental stage (two for each hive)). Immediately after sampling, the samples were snap frozen in liquid nitrogen and stored at −80 °C until further use.

2.2. RNA isolation

Larval RNA was isolated from pools of five individuals, RNA from adults and pupa was isolated from individuals and pooled likewise post isolation. The frozen samples were pulverized using a stainless-steel bead with four disruption cycles in a bead mill MM 300 (Retsch GmbH, Germany) at 20 Hz for 30 s. RNA from first instar larvae was isolated using the PicoPure isolation kit (ThermoFisher, USA) according to the manufacturer's protocol. RNA from older larvae was isolated using the RNeasy Lipid Tissue Mini Kit (Qiagen, Germany). An on-column RNase-Free DNase (Qiagen, Germany) digest was included in both isolation procedures. Disrupted pupal and adult tissue was lysed using TRIzol™ Reagent (Invitrogen, USA) and crude RNA was isolated using phenol-chloroform extraction. The RNA was further purified from the aqueous phase via magnetic beads using the Agencourt RNAdvance Tissue Kit (Beckman Coulter, USA). A DNase I digest (Agilent, USA) was included. RNA quality was assessed using the QIAxcel RNA QC Kit v2.0 and quantified using a Tecan M200 spectrophotometer (Tecan Group, Switzerland). High-quality total RNA was sent to Eurofins Genomics for sequencing (75 bp paired-end reads on an Illumina NextSeq500). Sequencing failed for one sample resulting in gene expression data for 47 different samples. The data sets have been uploaded to NCBI and can be accessed under bio-project number PRJNA728435.

2.3. Bioinformatics

Generated sequencing reads from all samples were aligned to the honey bee genome assembly Amel_4.5 (GCF_000002195.4) using the splice site information from the official gene set v 3.2 (OGS) (Elsik et al., 2014) via STAR 2.5.3ab (Dobin et al., 2013). Aligned reads were counted using subread grouping on the gene level (Liao et al., 2014). Genes were annotated using the official annotation data set from the Hymenoptera Genome Database (Elsik et al., 2016). In addition, the OGS coding sequences were BLASTed (blastx, evalue≤1e-3) vs. SwissProt (9966 hits) and vs. representative sequences from the GeneOntology resource (8811 hits) and queried against known protein motifs using InterPro (Paysan-Lafosse et al., 2023). Mapped read counts were analyzed for differential expression between groups using DESeq2 (Love et al., 2014). For visualization, the data were log2-transformed (after adding 0.5 to each value) and normalized further to a common median of 10 across all samples. To assess general distribution patterns and

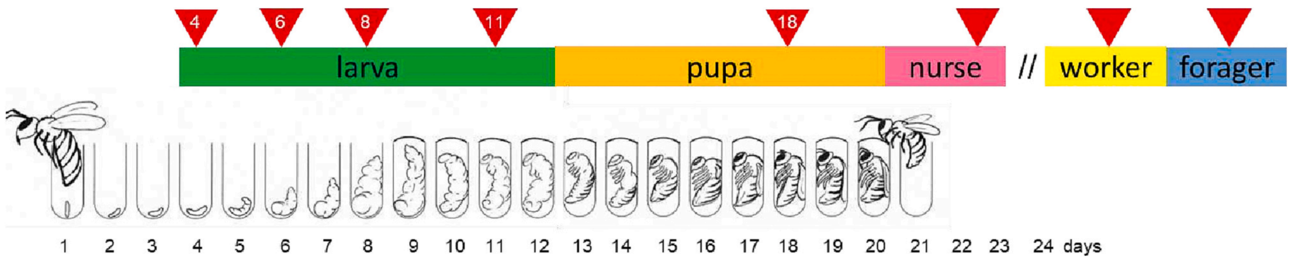

Fig. 1. Honey bee life cycle and sampling intervals for RNAseq analysis. Honey bee life stages were sampled at different time intervals after egg laying as indicated by red triangles. In total eight different life stages were collected including larvae (4d, 6d, 8d, 11d), pupae (18d) and adults, separated by collection sites (nurse: brood chamber; worker: honey super; forager: flight hole). Samples were collected from three different hives at two sampling dates each between late May and mid July. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

variability between samples a principal component analysis (PCA) was conducted.

To identify genes with differential expression between the different developmental stages N-way ANOVA was used with developmental stage and date of sampling as factors ($p < 0.05$). The identified genes were ranked based on their p-value and the top 1000 genes were used for an initial K-means clustering. Additionally, another K-means clustering of all genes was performed. For further analysis, clusters from k = 4 (top 1000 genes) and k = 5 (all genes) were chosen based on the formation of distinct expression profiles at these stages. For identification of over-represented GO terms in each cluster, a Fisher's exact test was performed. Heatmaps and boxplots with expression values of detoxification genes were created in R version 4.1.0 using the ggplot2 and reshape2 libraries.

2.4. Phylogenetic tree construction

Protein sequences were retrieved from the NCBI database and aligned using CLC Genomics Workbench 20.0.4 (Qiagen, Germany). Partial sequences were excluded. The phylogenetic trees were constructed via neighbor-joining method (Saitou and Nei, 1987) using Jukes Cantor as the protein distance measure and performing bootstrapping (n = 1000). The bootstrap threshold was set at 40%. Annotations for P450 genes were adopted from the official P450 nomenclature (Nelson, 2006). For CEs and GSTs, the classification used by Claudianos et al. (2006) was adopted. ABC transporters were classified based on (Dermauw and Van Leeuwen, 2014). Honeybee UGT protein sequences were submitted to the UGT nomenclature committee (https://labs.wsu.edu/ugt/) and classified accordingly (see supplement).

2.5. Tissue specific gene expression analysis in honeybee adults

Tissues were collected from pools of at least 10 adult honeybees sampled at three different time points (3 biological replicates). For dissection, bees were anesthetized with CO_2 and the head was cut off. The brain was isolated by opening the cranial vault and the compound eyes. Dissection of the midgut and Malpighian tubules was performed by opening the abdomen using conventional methods. Fat tissue located in the abdomen was collected, and the dissected cuticle was cleansed and used for RNA isolation. Tissue samples were directly snap frozen in liquid nitrogen and stored at −80 °C until further use. The frozen samples were ground with a stainless-steel bead with four cycles at 20 Hz for 30 s in a Mixer Mill MM 300 (Retsch GmbH, Haan, Germany). The ground tissue was lysed with TRIzol™ reagent (Invitrogen, Carlsbad, CA, USA). RNA was isolated using the RNeasy Mini Kit (Qiagen, Hilden, Germany, including a RNase-Free DNase digestion step).

RNA was quantified spectrophotometrically (NanoQuant Infinite 200, Tecan, Switzerland) and its integrity was checked by an automated gel electrophoresis system according to the CM-RNA and CL-RNA methods (QIAxcel RNA QC Kit v2.0, Qiagen, Hilden, Germany). For quantitative PCR (RT-qPCR), the iScript cDNA Synthesis Kit (Bio-Rad, Hercules, CA, USA) was used for cDNA generation, using 750 ng of RNA per reverse transcription. Real-time PCR was performed in triplicate using SsoAdvanced Universal SYBR Green Supermix (Bio-Rad, Hercules, CA, USA) with 2.5 ng of cDNA and 0.25 μM of each primer (Table S1) in a total reaction volume of 10 μL with a CFX384™ Real-Time System (Bio-Rad) and non-template mixtures as negative controls. The PCR program was as follows: 95 °C for 3 min; 95 °C for 15 s; 60 °C for 15 s; 60 °C for 15 s, reading off the plate; steps 2–5 were repeated 30 times, followed by a final post-PCR melt curve (ramping from 65 °C to 95 °C by 0.5 °C every 5 s) to check for nonspecific amplification. Amplification efficiency was determined for each primer pair, and inter-run controls were performed in each run to minimize plate/run-specific effects. Two reference genes, ubiquitin carboxyl-terminal hydrolase CYLD (GenBank: LOC410344) and TBC1 domain family member 20 (GenBank: LOC100578721), were selected for normalization. These were previously determined and showed good stability across tissues (CYLD: M 0.598; CV 0.193; TBC1: M 0.598; CV 0.220). The commonly used reference genes GADPH and RPS18 showed less stable GeNorm M and Coefficient of Variation values across the different tissues than the two selected genes. These were 1.44 (M) and 0.635 (CV) for GADPH and 1.340 (M) and 0.527 (CV) for RPS18, respectively. In selecting the reference genes, we ensured that the values did not exceed 1.000 for the GeNorm M or 0.500 for the coefficient of variation (CV) as recommended by qbase + software version 3.3 (Biogazelle, Zwijnaarde, Belgium) which was used for gene expression analysis. Gene expression analysis was performed using qbase + software version 3.3 (Biogazelle, Zwijnaarde, Belgium) (Hellemans et al., 2007). Mean normalized expression values were converted into logarithmic data and heat maps were generated using the Morpheus software (https://software.broadinstitute.org/morpheus; version: December 2022).

3. Results

3.1. Transcriptomic differentiation of different developmental stages of the honeybee

RNAseq was performed from a total of 47 different samples (8 different life stages, 3 hives, 2 time points for each hive; one sample (6 DAO) failed during sequencing) and reads per sample ranged from ~4.9 million to ~12.6 million. Average mapping rate to the honeybee genome was 89.3% resulting in an average coverage of 44× per gene. On average, 78.2% of the transcripts from the official genome assembly were considered expressed based on expression values >0 transcripts per million (tpm). Principal component analysis (PCA) revealed a clear distinction of overall gene expression between the different developmental stages (adults, pupae, and larvae sampled at different days after oviposition (DAO)) with 38.7% and 13.8% of the variability explained by two principal components (Fig. 2). A clear separation is also seen between the different larval stages. Larvae entering the pupal stage and sampled at 11 DAO clustered close to pupae, suggesting the initiation of the pupal stage. Contrary, adults sampled from different hive locations

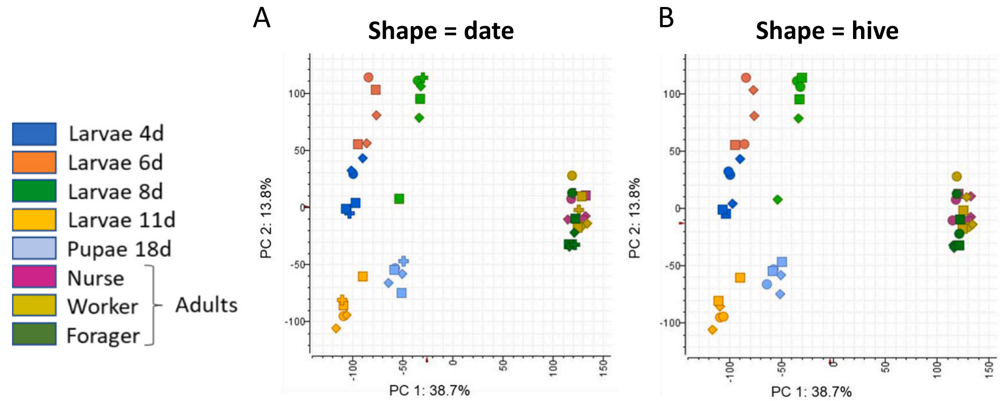

Fig. 2. Principal component analysis (PCA) and PC scores of the first two components. PCA of gene expression levels at larval stages, pupae and adults analyzed by (A) date and (B) hive. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

(brood chamber, honey super, flight hole) were not distinguished based on their gene expression profile. In general, the gene expression pattern was largely independent from the date of collection (Fig. 2A) or the hive (Fig. 2B), indicating a low variability of spatiotemporal gene expression patterns.

3.2. *Profiling of expression patterns of detoxification genes*

To identify genes following a similar expression profile across developmental stages, K-means clustering was performed. First, differentially expressed genes were identified via N-way ANOVA ($p < 0.05$). The top 1000 ranked genes yielded by ranking the result of N-way ANOVA (day, date)

by BH q-value ($p < 2.29\text{e-}10$, $q < 3.5\text{e-}9$) from this analysis were taken and partitioned into four clusters ($k = 4$) resulting in distinct expression patterns (Fig. 3A). Cluster 1 (271 out of 1000 genes) is characterized by a steady expression in larval stages which drops during the transition to pupa and remains flattish in adult bees. Genes in cluster 2 (356/1000) were expressed at similar levels in larval and pupal stages, while expression is highly elevated in adults. For cluster 3 (247/1000), a steady increase of expression was observed during the larval development followed by a sharp decline at the pupal stage and a steep increase to high expression levels in adults. Finally, cluster 4 (126/1000) is defined by a sharp increase in expression levels during the larval development with a subsequent decline to lower levels in pupae and adults.

Following the identification of distinct expression clusters, we queried each cluster for the presence of Gene Ontology (GO) domains to see if specific terms within a domain might be overrepresented in a specific cluster (Fig. 3B). None of the domains seem overrepresented in a specific gene expression cluster. However, specific GO terms related to enzyme families involved in environmental response and xenobiotic detoxification were overrepresented in cluster 2 and 3. In cluster 2, the terms "monooxygenase activity" (P450), "transmembrane transporter activity" (ABC) and carboxylic ester hydrolase activity" (CCE) were among those significantly overrepresented (Table S2). Similarly, in cluster 3, "monooxygenase activity", "transporter activity", and "glucuronosyltransferase activity" (UGT) were among the overrepresented

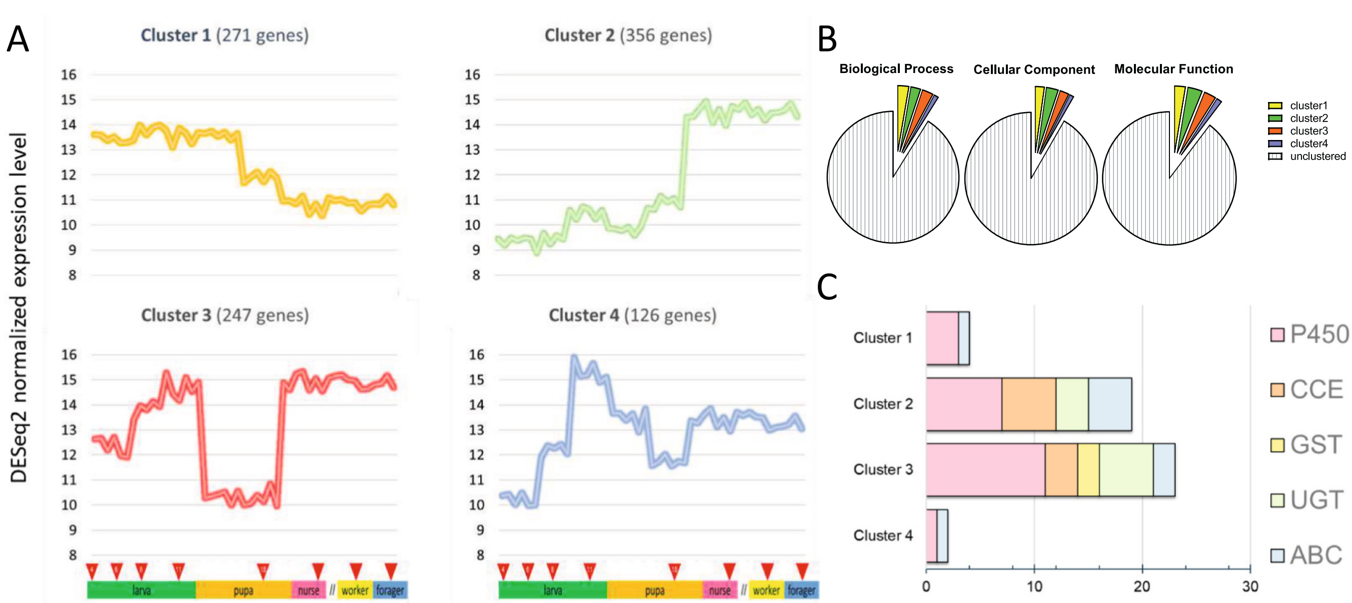

Fig. 3. K-means clustering of genes following similar expression profiles across honey bee life stages. (A) K-means clustering of the top 1000 genes yielded by ranking the result of N-way ANOVA (day, date) by Benjamini-Hochberg q-value ($p < 2.29\text{e-}10$, $q < 3.5\text{e-}9$). DESeq2 normalized expression levels are centered on a common median of 10. (B) Proportion (%) of the three domain ontology domains in each cluster. (C) Number of detoxification genes found in each cluster based on a total number of 133 previously described honey bee detoxification genes (Berenbaum and Johnson, 2015). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

terms (Table S3). Assessing the cluster-specific distribution of the major detoxification enzyme families (P450s, CCEs, GSTs, UGTs and ABCs) confirmed a strong accumulation of these genes in cluster 2 and 3, dominated by P450 genes followed by CCEs, ABCs and UGTs in cluster 2 and UGTs, CCEs, ABCs and GSTs in cluster 3, respectively (Fig. 3C; Table S4). These results revealed that several candidate genes from major detoxification enzyme families are particularly abundant in adults, and incrementally upregulated during larval development, while their transcription is downregulated in pupal stages, likely indicating an involvement in environmental response and xenobiotic clearance in mobile life stages.

Next, we repeated the K-means clustering analysis (k = 5) with all genes in order to get a more holistic overview on transcript levels following a similar expression pattern across life stages (Fig. S1). Cluster 3 of this approach, encompassing 2553 genes, followed a similar gene expression pattern across life stages as obtained for K-means cluster analysis of the top 1000 genes. Genes of this cluster are characterized by a moderate, but steady increase in expression during larval development, a significant drop in expression during the pupal phase and highest levels of expression upon pupal eclosion in adult honeybees (Fig. 4A).

Since the expression profile of genes represented in cluster 3 follows the pattern one would expect from enzymes involved in xenobiotic detoxification, we focused our analysis on this cluster. Indeed, half of the genes of the entire honeybee detoxification gene inventory (65 out of 133 (as reviewed in Berenbaum and Johnson, 2015)) followed the expression profile represented by K-means cluster 3 (Fig. 4B), including 29 P450s, followed by ABCs ($n = 15$), CCEs ($n = 13$), UGTs ($n = 6$) and GSTs ($n = 2$) (Table S5). Interestingly, out of the 29 P450 genes in cluster 3, 24 belong to clan CYP3 (Table S5). In contrast, none of the mito clan P450s was found in cluster 3 but in three other clusters (Table S6). However, it must be noted that the depicted expression pattern for cluster 3 (Fig. 4A) represents the median of all genes in this cluster, so that the expression pattern of each individual gene may slightly deviate from this. But the general tendency is shared for all genes in a cluster as shown exemplary in detail by heatmaps of gene expression levels for 47 P450 genes across life-stages (Fig. 5). For expression levels of individual genes of all other detoxification gene families please refer to Figs. S2–4.

The observed similarity between temporal expression patterns of more than half of the detoxification gene inventory in cluster 3 encouraged us to check which of the commonly known transcription factors followed the same expression pattern. Interrogating cluster 3 for the presence of genes assigned to the GO-term “DNA-binding transcription factor activity” (GO:0007300) resulted in 16 hits (Table S7), including several proteins previously described in other insect species to be involved in detoxification gene regulation such as “cap ‘n’ collar” (CNC), Maf or CREB (Amezian et al., 2021). A more detailed view on the expression pattern across life stages of some selected transcription factors is provided in Fig. S5.

3.3. Phylogeny of detoxification gene families

Next, phylogenetic analysis of the five major detoxification gene superfamilies was performed to investigate the evolutionary relationships between genes and identify potential branches which appear to be co-regulated regarding their expression profile across the honeybee life cycle. The honeybee CYPome is divided into four clans: clan CYP2, CYP3, CYP4 and CYPmito. Predominant in cluster 3 is clan CYP3 with 24 P450s (Fig. 6A). In addition, three clan CYP4 genes as well as two clan CYP2 genes are present in cluster 3 while none of the mitochondrial clan P450s followed the described pattern. For the second enzyme family involved in phase I metabolism (CCEs), the phylogenetic analysis was more ambiguous with bootstrap support dropping below the threshold of 40% for several genes (Fig. 6B). Nonetheless, a clear separation is seen within the tree regarding the presence of genes in cluster 3. Genes comprising the neuro / developmental class of CCE (as defined by Claudianos et al. (2006)) were not present while all genes classified as hormone / semiochemical processing class, and 7 out of 8 genes belonging to classes involved in dietary and detoxification processes are part of cluster 3. Phase II metabolism families UGTs and GSTs are characterized by a low number of genes with 11 and 9 genes present in our transcriptomic analysis, respectively. For UGTs, cluster 3-like genes are the majority with 6 out of 11 genes present in several families with no apparent pattern regarding their evolutionary origin (Fig. 6C). For GSTs, only 2 genes are member of cluster 3: one gene of the sigma class and one gene from the delta class (Fig. 6D). Next to P450 genes, the ABC transporters are most represented in cluster 3. ABC transporter genes are divided into subfamilies A to H. Predominant in cluster 3 are members from the ABC-C subfamily (7 out of 9), followed by genes out of the subfamilies D, G, H, A and B (Fig. 6E).

3.4. Tissue-specific expression of potential detoxification genes

As described above, K-means clustering analysis revealed an abundance of detoxification genes following a similar expression pattern (K-means cluster 3), i.e., a steady increase in growing larval instars, followed by a significant drop in pupae and a steep upregulation in adults. Such an expression pattern is supportive for an involvement of these detoxification genes in xenobiotic defense; however, their upregulation and presence in tissues involved in xenobiotic detoxification would provide a further line of evidence for their role. Therefore, we examined the tissue-specific expression pattern of 22 candidate detoxification genes resulting from K-means clustering (k = 4, top 1000 genes yielded by ranking the result of N-way ANOVA (day, date) by BH q-value ($p <$ 2.29e-10, $q <$ 3.5e-9)) and represented in cluster 3 (Table S4). The results obtained by RT-qPCR for the expression level of these candidate

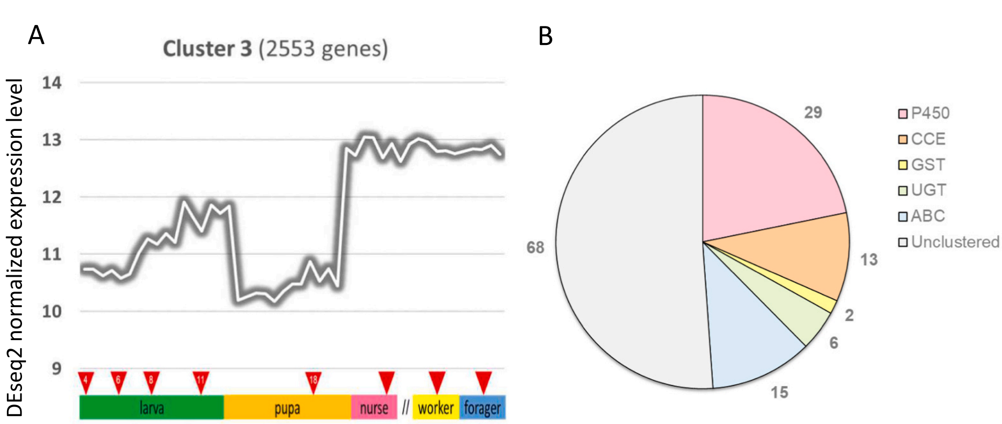

Fig. 4. K-means clustering analysis. (A) Genes showing a similar expression profile across honey bee life stages and resulting from k-means clustering of all genes (cluster 3, k = 5). DESeq2 normalized expression levels are centered on a common median of 10. (B) Number of detoxification genes of each family found in cluster 3 based on a total number of 133 previously described honey bee detoxification genes (Berenbaum and Johnson, 2015), including 46 P450s, 24 CCEs, 10 GSTs, 12 UGTs and 41 ABCs. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

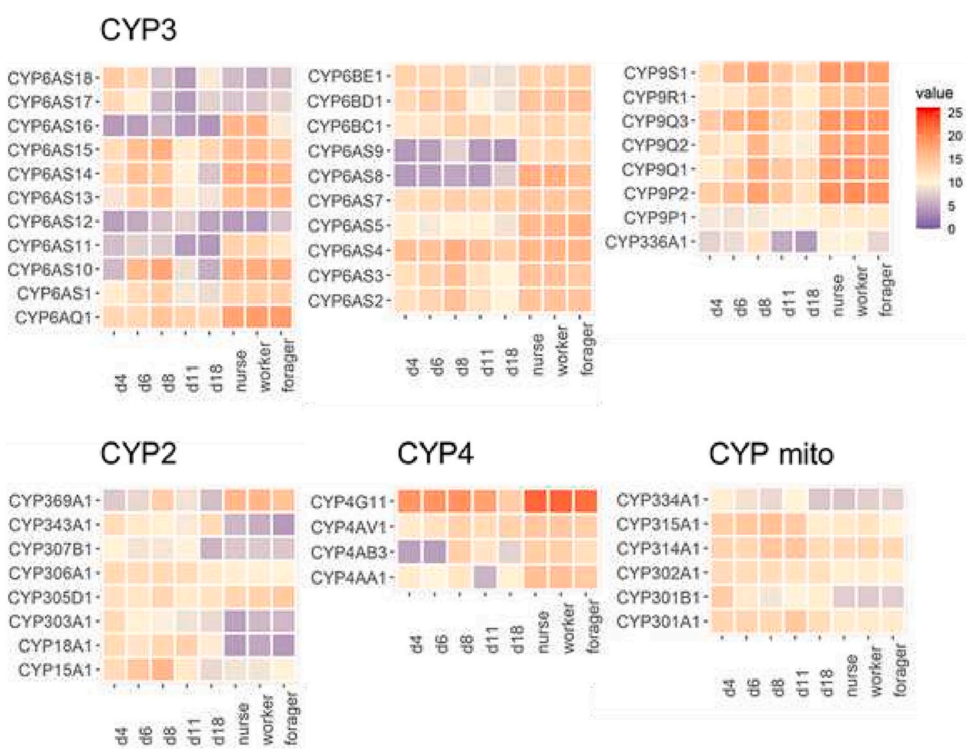

Fig. 5. Expression profile of honey bee cytochrome P450 (P450) gene inventory. Heatmaps of gene expression levels across life-stages, data were DESeq2 normalized and log2 transformed (after adding 0.5 to each value). These values were further normalized to a common median of 10 across all samples. Abbreviations: d4 = day 4 after oviposition, d6 = 6 days after oviposition, etc. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

genes in brain, midgut, Malpighian tubules, fat body and cuticle of adult worker bees are shown in Fig. 7. Overall, the Malpighian tubules were the predominant expression site for most of the analyzed candidate genes. For the CYP6AS family some differences in tissue-specific expression were noted, e.g., CYP6AS10 seems to be expressed predominantly in the midgut, whereas CYP6AS14 is the only gene lacking significant expression in the Malpighian tubules. Instead, its expression is highest in fat body and cuticle. For CYP9Q2 and CYP9Q3 we confirmed Malpighian tubules as the primary expression site followed by brain tissue. Genes from the other detoxification enzyme families highly expressed in the Malpighian tubules, include for example the multidrug resistance protein 4 (MRP4)-like ABC transporters GB43189 and GB45277, but also GSTs, UGTs and CCEs (Fig. 6). However, some candidate genes such as GB47299 (CCE), GB44058 and GB47301 (both UGTs) exhibited high expression in the honeybee midgut.

4. Discussion

Honeybees are exposed to a wide range of chemicals in their environment, but how they cope and interact with the diversity of xenobiotics at the molecular level is still poorly understood. For most cases where the molecular determinants of tolerance towards toxins of natural or synthetic origin have been identified, P450-mediated detoxification plays a key role (Katsavou et al., 2022). Other parts of the detoxification machinery have been studied less frequently. The data obtained here strongly support the notion that the expression levels of many P450s follow a temporal pattern, likely rendering them critical for the detoxification of foreign compounds in honeybees. P450s are the prevalent detoxification enzyme family in feeding honeybee life stages undergoing incremental upregulation in developing larvae until pupation and showing a steep increase again in expression levels in adult stages. K-means clustering of differential gene expression across life stages revealed that many honeybee genes, including more than half of the honeybee P450 gene inventory, follow such a pattern as observed in K-means cluster 3 of our analysis. Other studies previously speculated that such a pattern is more likely for P450s involved in xenobiotic detoxification (Chung et al., 2009). Indeed, approx. 80% of the CYP3 clan P450s represented by CYP6 and CYP9 family genes (and *CYP336A1*) follow this expression pattern. This supports our assumption that environmental response genes may be found predominantly in this expression cluster, indicating their involvement in the elimination of xenobiotics encountered by active life stages. In fact CYP3 clan P450s have been previously well characterized for their detoxification capacity against a variety of foreign compounds in honeybees, including neurotoxic insecticides with rather low acute toxicity to honeybees such as the diamide chlorantraniliprole, the butenolide flupyradifurone, the neonicotinoid thiacloprid, the pyrethroid *tau*-fluvalinate and the organophosphate coumaphos, whereas other members of these chemical classes are much less tolerated in terms of acute toxicity (Haas et al., 2021a, 2021b; Manjon et al., 2018; Mao et al., 2011, 2009). P450s of the CYP3 clan are close paralogs which evolved from lineage-specific family expansions and known to mediate xenobiotic metabolism and pesticide resistance (Feyereisen, 2011; Nauen et al., 2022). Interestingly, also three out of four CYP4 genes follow the same expression pattern as well as two out of eight CYP2 clan genes, *CYP305D1* and *CYP369A1*. This raises the question of their specific functions, which remains largely nebulous. *CYP305D1* has been shown to be highly expressed in mandibular glands of worker bees (Wu et al., 2017), and a possible function as a first line of defense in salivary excretions while processing food remains to be investigated. Another study has shown that *CYP305D1* was induced upon exposure of honeybee adults to the neonicotinoid insecticide thiacloprid, but its functional expression did not reveal any capacity to degrade thiacloprid (Alptekin et al., 2016). An orthologous gene of *CYP369A1* from the parasitoid wasp *Meteorus pulchricornis* has been implicated in insecticide tolerance via RNAi studies (Xing et al., 2021), but direct, functional studies underpinning its role in detoxification are lacking. Out of the three CYP4 genes following a similar expression pattern, CYP4G11 has been shown, by functional expression, to serve a dual function in honeybees: as a decarbonylase for cuticular hydrocarbon production and an odorant clearance enzyme in antennae (Calla et al., 2018). P450s of the insect-specific CYP4G subfamily have been shown to play a key role in cuticular hydrocarbon synthesis and expression in the same tissues throughout insect development (Balabanidou et al., 2016; Chung et al., 2009). Notably, a

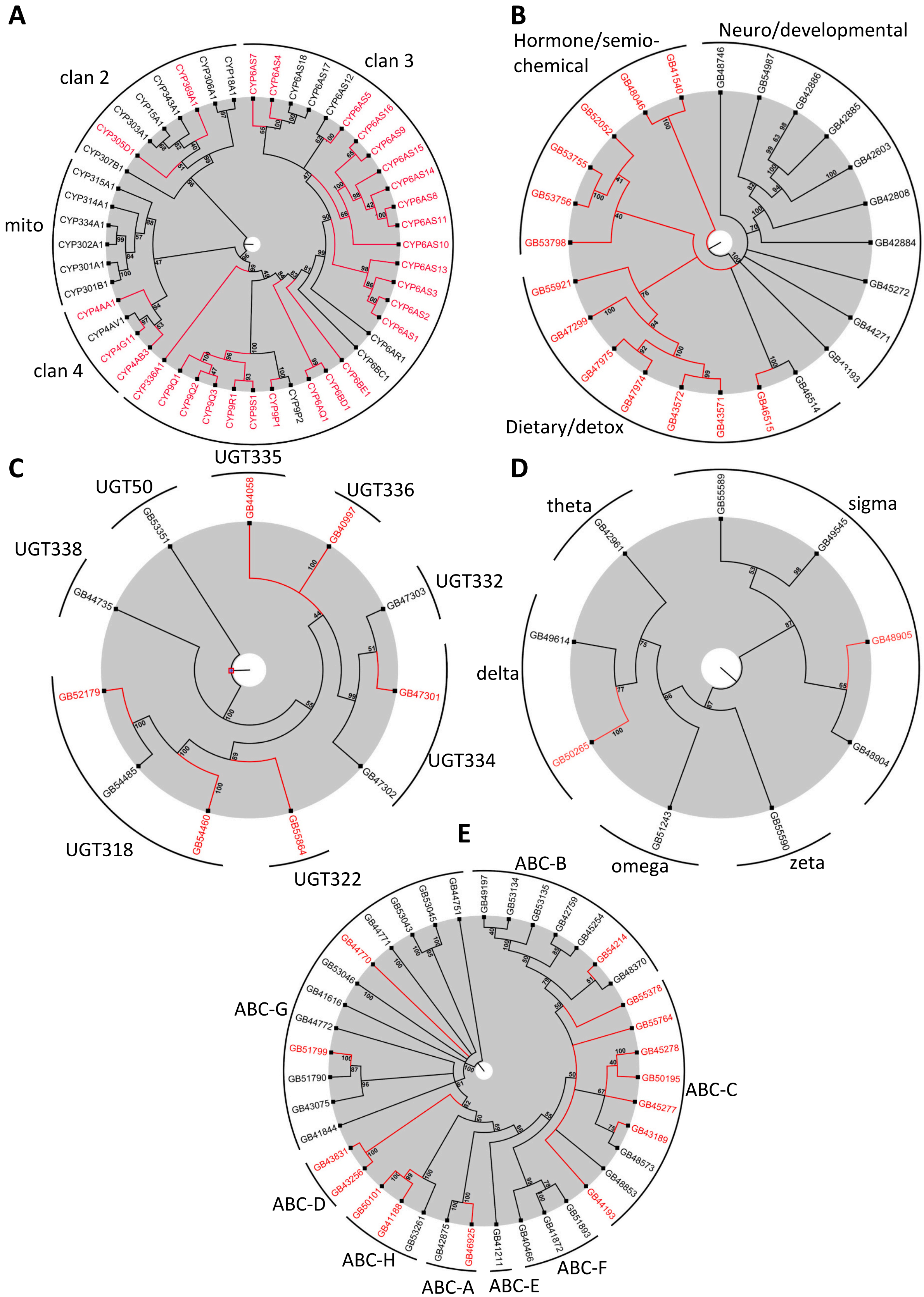

Fig. 6. Phylogenetic trees of detox enzyme families. (A) cytochrome P450 monooxygenases (B) Carboxylesterases (C) UDP-glycosyltransferases (D) Glutathione S transferases (E) ABC transporters. Bootstrap threshold was set at 40%. Those genes following the same expression profile in cluster 3 (k-means (k = 5), clustering of all genes) are highlighted in red. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

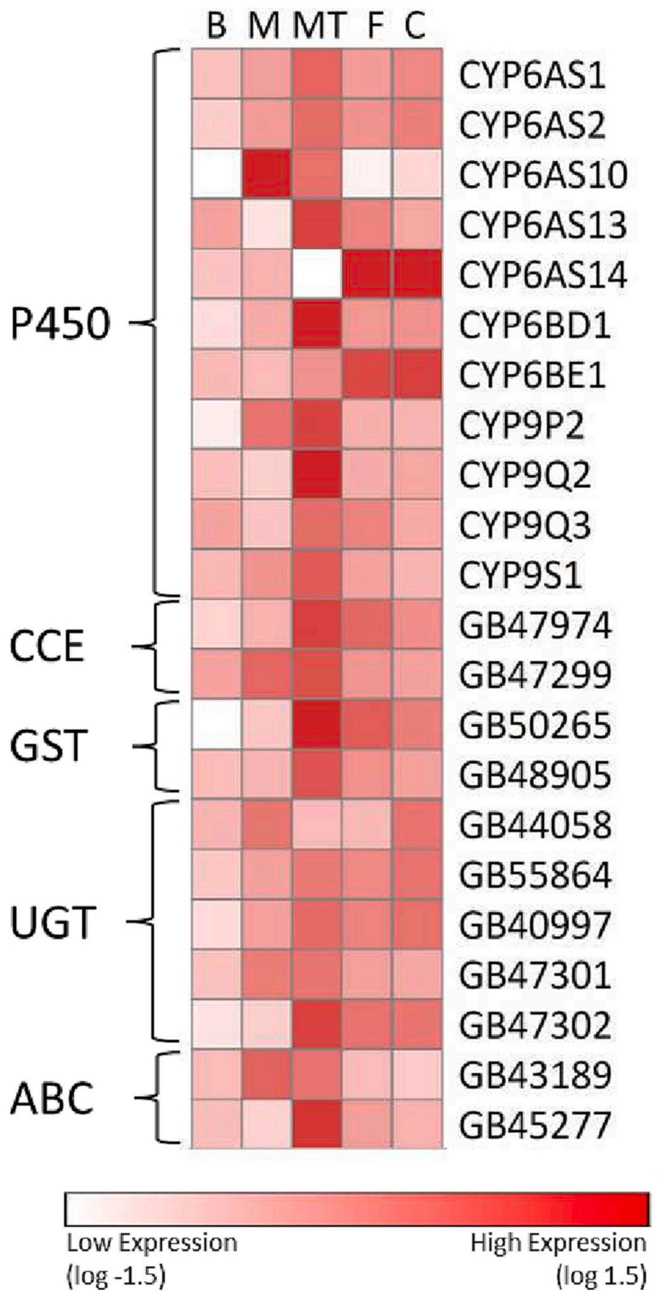

Fig. 7. RT-qPCR analysis of the expression level of detoxification genes. Heatmap highlighting the normalized expression level in different tissues of 22 detoxification genes assigned to cluster 3 (k-means clustering of top 1000 genes, k = 4; Table S4). Abbreviations: B, brain; M, midgut; MT, Malpighian tubules; F, fat body; C, cuticle. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

previous study revealed that *CYP4G11* was rapidly upregulated in *Apis cerana* exposed to different stressors, including pesticides (Shi et al., 2013). In the present study, *CYP4G11* was one of the *CYP* genes showing the highest expression level across honeybee life stages, including pupae. A similar pattern of expression has been described for its ortholog in *A. cerana* (Shi et al., 2013). This example highlights that following a certain expression pattern is not sufficient evidence for a particular function, but merely a starting point for subsequent characterization of most relevant candidate genes. However, candidate CYP2 and CYP4 genes highlighted above may deserve further attention regarding their detoxification capacity towards xenobiotics in the future.

Phase I metabolism is often rate-limiting in the context of acute insecticide and natural product toxicity (Yu, 2011). As such, members of CCE enzyme family are often a critical determinant of insecticide toxicity in insects, particularly for organophosphates and pyrethroids (Wheelock et al., 2005). Indeed, it has been shown that ester cleavage of pyrethroid insecticides is also a relevant detoxification mechanism in honeybees (Johnson et al., 2006). However, studies on specific esterase genes and their potential involvement in xenobiotic response in honeybees are lacking. Two studies reported the induction of a single esterase gene in honeybee adults exposed to the organophosphate coumaphos (Schmehl et al., 2014), and the neonicotinoid thiacloprid respectively (Alptekin et al., 2016). Therefore, the present study provides a starting point for future research on this detoxification enzyme family in honeybees, because especially CCE genes from the dietary/detox and hormone/semiochemical processing class followed the expression profile of K-means cluster 3.

While the functional verification of the involvement of phase I enzymes such as P450s in xenobiotic - particularly insecticide - detoxification is relatively straightforward to assess, the role of phase II enzymes such as GSTs and UGTs in conjugating functionalized substrates can be more difficult to elucidate, because appropriate substrates for in vitro studies are lacking.

GSTs are well-known for their role in insecticide resistance and the detoxification of xenobiotic compounds (Koirala et al., 2022). They were shown to play a dual role in the metabolic fate of insecticides by facilitating phase II conjugation and direct metabolization of insecticides by reductive dehydrochlorination (e.g., DDT) or sequestration (e.g., pyrethroids) (Kostaropoulos et al., 2001; Riveron et al., 2014). Often however, their primary substrate is not the parent compound itself but rather a hydroxylated phase I metabolite as demonstrated for cyflumetofen (Pavlidi et al., 2017). Additionally, GSTs have been shown to reduce insecticide toxicity by exhibiting peroxidase activity, thus offering an indirect resistance mechanism by reducing potential oxidative stress caused by insecticide exposure (Abdollahi et al., 2004; Vontas et al., 2001). The potential functional role of GSTs in xenobiotic defense in honeybees remains largely elusive, and a recent study showing the GST-mediated reduction of oxidative stress after insecticide exposure is the only example described in bees to the best of our knowledge (Shan et al., 2022). Here, we identified two interesting candidates from the delta and sigma GST classes which warrant further studies to elucidate their impact on insecticide toxicology, simply based on their temporal expression pattern and transcript levels in Malpighian tubules and fatbody, because members from both GST classes have been implied in insecticide resistance in other insects (He et al., 2018; Hu et al., 2020).

UGTs comprise another large family of phase II metabolic enzymes, and similar to GSTs they often work in concert with other metabolic enzymes (Ahn et al., 2012). Honeybees possess a rather small number of UGTs (12) compared to other insects such as *D. melanogaster*, where the spatiotemporal expression of all 35 UGT genes has been described in detail (Ahn and Marygold, 2021). Functional studies with insect UGTs involved in the detoxification of exogenous substrates are scarce, but there is evidence for their ability to conjugate plant defense chemicals in insects (Daimon et al., 2010; Israni et al., 2020). However, their relevance for insecticide resistance and especially the mechanistic implication of their involvement is less well understood. Increased insecticide susceptibility after RNAi-mediated silencing of candidate UGTs has been reported multiple times, but analytical verification of a glycosyl-conjugation of an insecticide by insect UGTs is lacking. More recent studies revealed that overexpressed UGTs in *Myzus persicae* and *Aphis gossypii* are likely involved in the sequential metabolism of sulfoxaflor and cyantraniliprole, respectively (Pym et al., 2022; Zeng et al., 2021). Functional validation was provided by the ectopic expression of the respective UGT resulting in low, but significant insecticide resistance in transgenic *D. melanogaster* when compared to wildtype flies. In honeybees UGTs have not yet been functionally shown to confer insecticide selectivity, but a recent study revealed upregulation of almost all UGTs in *A. cerana* upon exposure to insecticides (Cui et al., 2020). The highest level of upregulation was detected for *AccUGT2B20-like* and a role in overcoming oxidative stress has been hypothesized based on activity measurements of antioxidant enzymes after RNAi-mediated silencing of *AccUGT2B20-like*. The expression patterns of UGTs in this study revealed six promising candidates for future studies on their potential role in xenobiotic metabolism.

Next, we analyzed the expression pattern across life stages of 41 honeybee ABC transporters including those involved in phase III detoxification processes, i.e., shuttling a broad range of endogenous and xenobiotic compounds across cell membranes. Most ABC transporters are ATP-driven unidirectional efflux transporters and a recent phylogenetic study of the ABC superfamily across >150 arthropod species revealed specific expansions of ABC transporter families suggesting evolutionary adaptation congruent with a role in xenobiotic defense (Denecke et al., 2021). In the present study, ABC genes from six out of eight subfamilies mapped to K-means cluster 3 and followed an expression pattern suggesting a potential role in xenobiotic elimination in feeding honeybee life stages. Indeed, several transporters from various ABC subfamilies have been proposed to attenuate insecticide and plant secondary metabolite toxicity. For the ABCA family in arthropods little is known about their specific functions, but RNAi silencing of several members led to increased tefluthrin sensitivity in *T. castaneum* (Kalsi and Palli, 2017). Members of the ABCB family (often referred to as multidrug resistance proteins (MDRs) or permeability glycoproteins (P glycoproteins; P-gps) (Denecke et al., 2021)) are well-known for their involvement in drug resistance in mammalian cancer cells (Kartner et al., 1983), and have been implicated in cardenolide tolerance in lepidopteran species (Petschenka et al., 2013).Gene knockouts in Drosophila have linked several members of this family to insecticide tolerance (Denecke et al., 2017). ABCC subfamily members, also referred to as multidrug-resistance associated proteins (MRP), exhibit diverse functions including ion transport or cell-surface receptor activity (Dermauw and Van Leeuwen, 2014), but also mediate diflubenzuron and malathion tolerance in *T. castaneum* (Rösner and Merzendorfer, 2022, 2020). Like their mammalian counterparts, ABCC genes may prefer substrates conjugated by phase II enzymes which can complicate to proof their impact on insecticide elimination (Liu et al., 2012; Szeri et al., 2009). Nonetheless, the candidates of subfamily ABCA and ABCB and especially the seven ABCC genes present in cluster 3 are interesting candidates regarding toxin metabolism and tolerance in honeybees. In contrast, ABCD family members mapping to cluster 3 are most likely unrelated to xenobiotic metabolism. ABCD genes are conserved across taxa and transport acyl coenzyme A esters across the peroxisomal membrane (Morita and Imanaka, 2012; Wanders et al., 2007). Similarly, ABCE and ABCF genes are irrelevant for xenobiotic elimination as they lack transmembrane domains and thus the translocation function (Liu et al., 2011; Tyzack et al., 2000). Instead they are assumed to be important for ribosome biogenesis, translation control and mRNA output (Barthelme et al., 2011). The human ABCG family comprises 5 proteins with four of them having endogenous or dietary lipids (e.g. cholesterol) as their principal substrates (Kerr et al., 2011), whereas insect ABCGs are known for colour pigment transport, most notably *white*, *scarlet* and *brown* in Drosophila (Mackenzie et al., 1999). The *brown* orthologue of the honeybee (GB51799) shows very low expression levels in white larvae, while elevated in dark-eyed pupal and adult stages confirming its role in pigmentation processes rather than detoxification. ABCH genes are less well studied and little is known about their function as they are absent in mammals, plants and fungi (Wu et al., 2019). They may be involved in lipid barrier formation in the insect cuticle and development across life stages (Zuber et al., 2018). In summary, we identified several ABC genes from the subfamilies A, B and C as promising candidates for future investigations regarding their role in insecticide/pesticide selectivity in honeybees.

Finally, tissue-specific expression analysis of several genes across detoxification enzyme families supported their role as a defense mechanism against foreign compounds. Our analysis of selected candidate genes revealed Malpighian tubules as the predominant expression site in adult honeybees for many of these genes. During metamorphosis honeybee Malpighian tubules undergo major changes with four Malpighian tubules present in larvae being degraded during the transition into pupae to form 64 novel Malpighian tubules in adults (Gonçalves et al., 2018). Malpighian tubules are a predominant expression site for detoxification enzymes in insects (Yang et al., 2007), which has been confirmed for important detoxification enzymes in honeybees (Manjon et al., 2018), and other insects as well (Chung et al., 2009) supporting their active role in xenobiotic metabolism. Noteworthy is also the considerable expression of many of these genes in the honeybee brain which is the preferred site targeted by many insecticides. Detoxification genes expressed here may therefore play a critical role as the last line of defense against neurotoxic substances and deserve further characterization. Another aspect which warrants further research is the observed correlation of the spatiotemporal expression of candidate detoxification genes with those of several transcription factors potentially involved in the regulation of these genes. Indeed, knowledge on transcription factors and their functional role in honeybees is very limited (Guo et al., 2018). Our study provides starting points for future research, such as on Cap'n'collar (CNC) which has been called the “master regulator” of genes involved in xenobiotic detoxification in insects (Amezian et al., 2021).

In conclusion, the present study gives an overview about the differential expression of transcript levels of detoxification gene superfamilies across honeybee life stages. Several of these genes are potentially associated with insecticide metabolism and environmental response simply based on their temporal expression pattern, particularly restricted to those life stages actively feeding. The identification of genes sharing a specific spatiotemporal expression pattern will help to structure these enzyme superfamilies into different categories including xenobiotic defense. In combination with the identification of potential regulatory elements responsible for the expression of genes involved in insecticide/xenobiotic selectivity, this study provides a valuable resource of transcriptomic information. Not only for insect toxicologists, but also applied entomologists striving to assess pollinator health by considering molecular approaches to inform pesticide risk assessment (Haas and Nauen, 2021; López-Osorio and Wurm, 2020).

Authors contribution

Conceived, designed and supervised by RN. BL and GH collected life stages and prepared RNA. JR conducted RT-qPCR analysis. FM and JH carried out bioinformatic analyses. FM, JH, JR, and RN analyzed data and prepared figures and tables. JH, JR, and RN wrote the manuscript with contributions from all authors. All authors read and approved the final version of the manuscript.

Declaration of Competing Interest

The authors declare no competing interests.

Data availability

RNAseq data were deposited in GenBank under NCBI BioProject PRJNA728435. The bioinformatic analyses such as global gene expression data and annotations (as mapped to the honeybee genome) has been deposited in Mendeley Data (File: APISM_Atlas.xlsx) and accessible via this link doi:10.17632/7pyv3ccrzt.1. Honeybee UGT protein sequences were classified by the UGT nomenclature committee (https://labs.wsu.edu/ugt/) and an Excel file is attached in the supplementary information (UGT Annotation Committee_*Apis mellifera* UGTs_SI.xlsx).

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.pestbp.2023.105410.

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